Jorge Chavez

Joliet, IL | j.chavez3019@gmail.com | (815) 351-0478 | www.jorgechavezuiuc.com

EDUCATION

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

- Master of Science in Electrical and Computer Engineering - GPA: 3.83/4.0 — May 2025
  - *Researcher at the Assured and Trustworthy AI Research Lab (ASTRAL)*
  - *Concentration in Machine Learning Verification*
- Bachelor of Science in Computer Engineering - GPA: 3.80/4.0 — May 2023

PROJECTS

$\alpha\beta$-CROWN, Neural Network Verification – PyTorch Based Verifier — Jan 2024 – Present

- Contributed to the team's **1st-place finish** in two international neural network verification competitions (VNN-COMP 2024 & 2025), **securing the #1 rank on all 21 benchmarks**.
- Developed a novel framework that **reduced branch-and-bound subproblems by 64%** and enabled up to **85% faster verification**.
- Implemented a PGD-based soundness check integrated into the project's **CI/CD pipeline** to **automate validation of new contributions**, with support for local runs to streamline development.
- Optimized parallel execution in the branch-and-bound algorithm for complex queries, **halving runtime on several benchmarks**, and authored technical documentation for new and existing modules to improve maintainability for new contributors.

Autonomous Lane-Tracking – Python, ROS, Gazebo — Dec 2022 – May 2023

- Developed a full-stack autonomous driving pipeline for lane tracking and collision detection.
- Implemented homography-based perspective transforms and edge detection to perceive lane-markers from camera data.
- Designed and tuned a Stanley controller for closed-loop steering, integrating LiDAR for obstacle detection.
- Validated the system in Gazebo simulations and deployed it on a Polaris GEM vehicle for real-world testing on a **purpose-built 3.5 m-wide track**, completing a **loop segment of approximately ~39 m without lane departures.**

INTERNSHIP EXPERIENCE

Motorola Solutions | Embedded Systems Engineer — May 2022 – Aug 2022

Schaumburg, IL

- Developed a multi-threaded Python application for a custom software-defined radio (SDR) to monitor, classify, and replay radio signals as decoys, enhancing system security.
- Designed a memory-efficient signal processing pipeline using Welch's method**, reducing storage from 2 MB/min to 0.015 MB/min (~99% savings)**, enabling long-term online recording and prediction.
- Improved communication reliability by applying digital signal processing and predictive analytics for signal recurrence and pattern detection.

RSO INVOLVEMENT

Illini Space Society | Controls Programmer and Project Lead — Jan 2025 – May 2025

- Supervised a 3-member team to design and deploy feedback controllers for autonomous systems using PyTorch (NNs) and PyBullet (simulation).
- Applied both ML-based and classical approaches: engineered neural-network controllers with Lyapunov stability guarantees (verified via αβ-CROWN) for inverted pendulum/cart-pole systems, and implemented LQR to achieve stable quadrotor control at arbitrary equilibrium points.
- Delivered technical workshops on state-space modeling, feedback control, and Kalman filtering, mentoring ~15 peers and strengthening the org's expertise in control theory.

Research and Scholarly Work

- **Clip-and-Verify: Linear Constraint-Driven Domain Clipping for Accelerating Neural Network Verification**
  NeurIPS 2025 (accepted, equal contribution)
- **Guaranteed Bounding Meshes Extraction from Neural Implicit Surfaces via Neural Network Verification**
  ICLR 2026 (under review, equal contribution)
- **A Linear Constraint Driven Approach to Efficiently Enhancing Branch and Bound in Neural Network Verification**
  Master's Thesis, UIUC, 2025

SKILLS

Programming: C/C++, Python, Matlab, Verilog, CUDA**,** PyTorch, OpenCV, Pandas, ROS